

RL-based Network Slice Embedding over Space Division Multiplexed Elastic Optical Networks

Divya Khanure^{*}, Riti Gour[†], Congzhou Li^{*}, and Jason P. Jue^{*}

^{*}Department of Computer Science

The University of Texas at Dallas, Richardson, Texas 75080, USA

[†] Department of Aviation and Technology,

San Jose State University, San Jose, California 95192, USA

Email:{divya.khanure, congzhou.li, jjue}@utdallas.edu, riti.gour@sjsu.edu

Abstract—Network slicing over space-division-multiplexed elastic optical networks (SDM-EONs) requires jointly managing spectrum, spatial cores, and compute resources, a coupling that many existing studies ignore by treating compute placement independently from routing and spectrum decisions. This disconnect can cause the spectrum to be allocated along a path, only for the request to fail due to insufficient compute resources along the path, or may result in compute resources being allocated without consideration for spectrum resource availability on the path between compute nodes. We propose a path-constrained reinforcement learning framework that addresses compute node selection and RMCSA, being aware of both resources, restricting the RL agent’s action space to nodes along k -shortest paths between request endpoints. Training incorporates reward shaping to improve robustness under high load. We evaluate the proposed approach, that is, PPO-Full, which jointly selects compute nodes and routing paths via a multi-dimensional action space, against distance-based heuristics, a greedy baseline, and a decoupled VONE-DRL baseline on a 24-node USNET topology under hotspot traffic conditions. Results demonstrate consistent improvements in acceptance rate over all baselines at high load, with gains becoming more pronounced as traffic intensity increases.

Index Terms—Network Slicing, Slice Embedding, RMCSA, Reinforcement Learning, Elastic Optical Networks, Hotspot Traffic, Multi-dimensional Action Space.

Supporting high-bandwidth heterogeneous services over a shared optical infrastructure has driven growing interest in network slicing over space-division-multiplexed elastic optical networks (SDM-EONs), where slice embedding demands simultaneous management of spectrum, spatial cores, and compute resources, a joint dependency that many RMCSA formulations overlook [1]. In SDM-EONs, spectrum assignment must satisfy contiguity and continuity requirements, core selection must account for inter-core crosstalk, and compute placement must respect node capacity limitations, making joint optimization computationally intractable. Prior work has largely decoupled compute placement from routing and spectrum assignment, producing suboptimal mappings under dynamic conditions [2]–[4]. Furthermore, slice embedding studies enforce strict one-to-one virtual-to-substrate node mappings that limit compute utilization. Coordinated and co-location-aware strategies in broader VNE research have shown improved acceptance ratios [5], [6], but such principles

have not been widely incorporated into SDM-EON slicing frameworks.

Building on our prior heuristic-based work [7], which proposed Direct Path-Driven Slice Mapping (DPSM) and Waypoint-Assisted Multi-Segment Slice Mapping (WMSM), this paper incorporates reinforcement learning into the resource allocation process. DPSM assigns compute to the first feasible node along the path; its extended version targets a node near the path midpoint to balance segment lengths. WMSM decomposes requests into shorter independently provisioned segments via compute-capable waypoints; WMSM-F extends this by jointly minimizing path length and segment imbalance. While these heuristics perform well, their deterministic rules cannot adapt to dynamic traffic, motivating the present work, which retains WMSM-F as a greedy baseline and replaces fixed allocation logic with a learned RL policy.

We propose PPO-Full, a PPO-based approach for slice embedding in SDM-EONs that jointly optimizes compute node selection and RMCSA by restricting the agent’s action space to nodes along k -shortest paths, making allocation decisions aware of both compute and spectrum resources simultaneously. PPO-Full is evaluated against distance-based heuristics, WMSM-F, and a decoupled VONE-DRL baseline [8] on a 24-node USNET topology under hotspot traffic conditions. The main contributions are:

- A PPO-based slice embedding approach (PPO-Full) that jointly optimizes compute node and routing path selection via `MultiDiscrete([3, 3])`, coupling compute placement with RMCSA within a single learned policy. A node-only reranking variant (PPO-Rank) with path fixed to $k = 3$ was also explored as an ablation to quantify the value of path selection freedom, confirming that joint node-path selection provides measurable benefit.
- A candidate-aware observation space that encodes per-candidate compute availability and segment spectrum state, enabling the agent to make informed joint compute-spectrum placement decisions across $K = 3$ candidate nodes and paths selected empirically after sweeping $K \in \{2, 3, 4, 5\}$.
- A consecutive co-location scheme for VNF placement,

where multiple consecutive VNF demands $\{f_i, \dots, f_j\}$ may be co-located on a single physical compute node, and all valid consecutive partitions of F VNFs across $1-F$ nodes are tried exhaustively. A quantitative analysis of co-location impact is deferred to an extended journal version due to page constraints.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model. Section IV describes the proposed framework. Section V presents numerical results. Section VI concludes the paper.

I. BACKGROUND LITERATURE

Early VNE formulations by Gong and Zhu [3] and Wang and Hu [2] improved spectral efficiency but were limited to single-core architectures without compute-aware placement. Multi-core fiber introduced RMCSA research with IC-XT constraints: Zhang *et al.* [9], Jin *et al.* [10], and Tang *et al.* [11] proposed crosstalk-aware SDM-EON embedding, while Heera *et al.* [12], [13] introduced congestion-aware RMCSA heuristics, that is, all treating compute placement as independent from optical allocation.

RL has emerged as a promising alternative for optical resource allocation. Chen *et al.* [14] proposed DeepRMSA for RMSA in single-core EONs, Pinto-Ríos *et al.* [15] extended DRL to MCF-EONs, and Xu *et al.* [16] combined GCN with deep RL for topology-aware RMSA. However, none incorporate compute placement, limiting applicability to joint compute-spectrum slice embedding.

On the VNE side, Wang *et al.* [5] and Chowdhury *et al.* [6] proposed co-location and coordinated node-link mapping strategies. Gu *et al.* [4] applied deep RL to RAN slice migration at higher architectural layers, while Doherty *et al.* [8] applied masked DRL to VONE over single-core EONs with decoupled agents, which we adapt as a baseline. Our prior work [7] addressed heuristic-based SDM-EON slice embedding with compute-spectrum coordination, the present work replaces deterministic heuristics with a path-constrained RL agent, as summarized in Table I.

TABLE I
COMPARISON OF RELATED WORKS BY SDM-EON, COORDINATION,
CO-LOCATION, AND SOLUTION TYPE

Reference(s)	SDM-EON	Coord.	Co-loc.	Solution Type(s)
[10]–[13], [17], [18]	✓	✗	✗	ILP, Heuristic
[4], [6]	✗	✗	✓	RL, Heuristic
[5]	✗	✓	✓	Heuristic
[2], [3]	✗	✗	✗	ILP, Heuristic
[14]–[16]	✗	✗	✗	RL
[8]	✗	✗	✗	RL
[7]	✓	✓	✓	Heuristic
Proposed	✓	✓	✓	Heuristic + RL

II. SYSTEM MODEL AND PROBLEM FORMULATION

We model an SDM-EON as a graph $G = (V, E)$, where each link supports $|\Phi_e|$ spatial cores with f spectrum slots each, and a subset $V_c \subseteq V$ of compute-capable nodes with finite capacity. Cores are grouped by spatial adjacency to mitigate inter-core crosstalk.

A. Definitions and Operational Semantics

Compute & OEO Semantics. Compute capacity is expressed in normalized units representing VNF execution. Each request places $|F|$ VNFs with independent demands, co-located or distributed across path nodes. Deploying a VNF triggers OEO conversion, terminating the incoming lightpath, and relaxing spectrum continuity to within each segment only, allowing decomposition into shorter independently provisioned parts.

Each request $r = (s, d, b_{req}, F)$, where $F = \{f_1, f_2, \dots, f_n\}$ is accepted when VNFs are placed at feasible compute nodes and spectrum is provisioned on all resulting segments, with routing, spectrum, and compute decisions coupled sequentially.

B. Cost Model

The average provisioning cost per accepted VNR is defined as:

$$C_{r_i} = \frac{\sum_{v \in V_c} c_{v,r_i} \mathbb{C}_{compute} + \sum_{e \in E} \gamma_{e,r_i} \mathbb{C}_{spectrum}}{\text{accepted VNRs}}, \quad (1)$$

where c_{v,r_i} is the sum of compute units actually allocated across all VNFs at node v , γ_{e,r_i} is the total spectrum slots used across all RMCSA path segments for request r_i , and $\mathbb{C}_{compute} = \mathbb{C}_{spectrum} = 1.0$ (equal weighting). This metric captures the average total resources, that is, compute units plus spectrum slots, consumed per successfully provisioned VNR. A lower value indicates more resource-efficient allocation, a method that accepts the same requests but routes them more efficiently will incur lower cost. The overall objective is to minimize this average cost while maximizing the acceptance rate.

C. Constraints

The slice mapping must satisfy the following constraints, else the request is blocked:

- **(C1) Modulation reach:** The total path length must satisfy $\sum_{e \in P_{r_i}^k} D_e \leq D_{\max}(M_{r_i}^k)$, $\forall r_i, \forall k$, where P_{r_i} is the selected path, D_e is the distance of link e , and $D_{\max}(M_{r_i})$ is the maximum transmission reach for modulation format M_{r_i} . Three modulation formats are supported: 16-QAM (≤ 500 km, 1 slot/Gbps), QPSK (≤ 1000 km, 2 slots/Gbps), and BPSK (≤ 2000 km, 4 slots/Gbps).
- **(C2) Spectrum contiguity:** Each request r_i occupies a contiguous slot set $\Gamma_{r_i,e,\phi} = \{s_{r_i}, s_{r_i}+1, \dots, s_{r_i}+f_{r_i}-1\}$, $\forall r_i, \forall e \in P_{r_i}^k$, where $f_{r_i} = b_{req} \times \eta(M_{r_i})$ is

the number of slots per Gbps for the chosen modulation format.

- **(C3) Spectrum continuity:** Identical contiguous slots are assigned on all links along a path: $\Gamma_{r_i, e, \phi} = \Gamma_{r_i, e', \phi}, \forall e, e' \in P_{r_i}$. Continuity across different segments k is not required due to OEO conversion at compute waypoints.
- **(C4) Compute feasibility:** $c_{v, r_i}^{(j)} \geq f_j, \forall r_i$, such that $\sum_{r_i} \text{left}(c_{v, r_i}^{(1)} + c_{v, r_i}^{(2)} + \dots + c_{v, r_i}^{(n)}) \leq C_v^{\text{comp}}, \forall v \in V_c$, where $c_{v, r_i}^{(j)}$ is the compute allocated to VNF j of request r_i at node v , and C_v^{comp} is the node capacity.
- **(C5) Core-group selection (crosstalk avoidance):** Each request is assigned a core $\phi_{r_i} \in \mathcal{G}_k$, where $\mathcal{G}_k$ denotes a non-adjacent core group (e.g., $\mathcal{G}_1 = \{1, 3, 5\}$, $\mathcal{G}_2 = \{2, 4, 6\}$, $\mathcal{G}_3 = \{7\}$), ensuring $\sum_k g_{r_i, k} = 1$, with $g_{r_i, k} = 1$ if $\mathcal{G}_k$ is used by r_i and 0 otherwise.

D. Performance Metrics

Acceptance rate measures the fraction of successfully provisioned requests. **Spectrum utilization** is $U = \frac{\sum_i \sum_e \gamma_{e, r_i}}{\mathbb{N}}$, where $\mathbb{N}$ is total network slots (links $\times$ cores $\times$ slots/core). **Provisioning cost** is the cumulative cost associated with compute units plus spectrum slots consumed. **Spectrum fragmentation** uses the external fragmentation index [19]: $F_{ex}(l, c) = 1 - B_{max}/F_{total}$, where B_{max} is the largest contiguous free block and F_{total} is the total free slots, averaged across cores as $F_{link}(l) = \frac{1}{C} \sum_{c=1}^C F_{ex}(l, c)$.

The slice embedding problem is NP-hard, generalizing RSA (reducible to wavelength assignment [20]) and VNE (equivalent to subgraph isomorphism [21]), with modulation reach, inter-core crosstalk, and compute placement, adding multidimensional bin-packing complexity [22], making exact optimization intractable and motivating the RL-based approach proposed here. PPO-Full's inference time per request ($\approx 135 \mu s$ for the policy forward pass on CPU) is comparable to the more complex heuristics such as Equidistant ($\approx 1,115 \mu s$) and VONE-DRL ($\approx 1,256 \mu s$), with the computational cost of training incurred offline only.

III. PROPOSED FRAMEWORK

PPO-Full is proposed for network slice embedding in SDMEONS, generating a small set of candidate compute nodes ranked by path distance and using a learned PPO policy to jointly select a compute node and a routing path. Restricting the candidate set to $K = 3$ nodes and $K = 3$ pre-computed shortest paths strikes a practical balance between decision quality and action space tractability: a larger candidate set exponentially increases the number of joint node-path combinations the agent must explore, leading to slower convergence and sparser rewards, while $K = 3$ retains sufficient diversity to cover the dominant trade-offs between compute proximity, path length, and spectrum availability. K was selected empirically after sweeping $K \in \{2, 3, 4, 5\}$, with $K = 3$ offering the best trade-off between candidate diversity and

action space tractability. PPO-Full jointly selects a compute node and a routing path via `MultiDiscrete([3, 3])`, yielding $K \times K = 9$ joint node-path combinations per decision step. A node-only variant (PPO-Rank) with path fixed to $k = 3$ was explored as an ablation, confirming that joint node-path selection provides measurable benefit over node reranking alone (results not included in the paper due to page limit). VNF demands are partitioned consecutively across compute nodes starting from the RL-selected anchor node, with remaining VNF groups greedily assigned to the nearest feasible nodes. PPO-Full is evaluated against distance-based heuristics, a greedy baseline, and a decoupled VONE-DRL baseline described at the end of this section.

A. Action Space

PPO-Full operates over $K = 3$ candidate compute nodes ranked by total path distance $d(s, n) + d(n, \text{dst})$ and filtered by compute feasibility, reducing node selection from $O(|V_c|)$ to $O(K)$. It uses `MultiDiscrete([3, 3])`, jointly selecting a compute node (dim 0) and a routing path (dim 1) from $K = 3$ pre-computed shortest paths, yielding $K \times K = 9$ joint node-path combinations per decision step, giving a total of $K^2 = 9$ discrete joint actions.

B. Observation Space

PPO-Full observes a normalized feature vector constructed per arrival event, comprising: request features ($s, d, b_{req}, f_1 \dots f_F$), network utilization, and network fragmentation ($2 + F + 2$ values), and per-candidate features, which are, compute availability ratio, normalized distances to source and destination, segment utilization, maximum free block, and fragmentation for each of the two resulting segments (9 values per candidate). For $K = 3$ candidates and $F = 4$ VNFs, this yields a $(2 + F + 2) + 3 \times 9 = 36$ -dimensional observation vector.

C. Reward Shaping

The reward function incentivizes acceptance while penalizing spectrum-inefficient allocations:

$$\mathcal{R} = \begin{cases} 5.0 + 2.0 \times \left(1 - \frac{\min(S, 200)}{200}\right) & \text{accepted} \\ -5.0 & \text{blocked} \end{cases} \quad (2)$$

where S is the total number of spectrum slots consumed by the accepted request. The acceptance reward is augmented by a slot-efficiency bonus that decreases linearly with spectrum usage, reaching zero for allocations consuming 200 or more slots. This encourages the agent to prefer compact, spectrum-efficient placements without explicitly penalizing fragmentation, simplifying the reward signal compared to multi-term formulations while retaining the incentive for resource-efficient provisioning.

D. PPO Agent and Hyperparameters

PPO-Full is implemented using PPO [23] via `stable-baselines3` with an MLP policy (`net_arch=[256, 256]`), well-suited for discrete action spaces under non-stationary traffic. Hyperparameters are summarized in Table II. The best checkpoint is selected based on mean acceptance rate evaluated every 50,000 steps at $\lambda \in \{15, 30, 50\}$.

TABLE II
PPO HYPERPARAMETERS

Hyperparameter	Value	Hyperparameter	Value
Learning rate	3×10^{-4}	Clip range	0.2
Batch size	256	Entropy coefficient	0.01
Steps per update	2048	Training timesteps	1,000,000
Epochs per update	10	Parallel environments	4
Discount γ	0.99	GAE λ	0.95

Algorithm 1 PPO-Full Allocation

Input: $(s, d, b_{req}, f_1 \dots f_F)$, $K = 3$ candidate nodes $\mathcal{C}$, $K = 3$ paths $\mathcal{P}_{sd}$, policy π_θ

Output: Allocation decision, spectrum and compute assignment

Build observation o from network state, request features, and per-candidate segment features

Query policy: $a \leftarrow \pi_\theta(o)$; decode a as (n_1, p) — anchor node and path index

Partition $\{f_1 \dots f_F\}$ consecutively from n_1 ;

if any VNF group cannot be placed **then**

return *Blocked*

end if

Decompose path: $s \rightarrow n_1 \rightarrow [n_2, \dots] \rightarrow d$

for each segment (u, w) **do**

 Select modulation M from $D(u, w)$; compute $n = b_{req} \times \eta(M)$

 Allocate n slots on (u, w) using core group order $\{0, 2, 4\} \rightarrow \{1, 3, 5\} \rightarrow \{6\}$

if fails **then**

return *Blocked*

end if

end for

Reserve compute units at all placed nodes; update state

return *Accepted* = 0

E. Baseline Approaches

All baselines use first-fit spectrum allocation across core groups, differing only in compute node selection.

1) *Distance-Based Greedy Heuristics*: Compute nodes are ranked by distance score relative to request endpoints using two strategies: **Endpoint-Biased** (proximity to source or destination) and **Equidistant** (minimizing $|d(s, v) - d(v, d)|$ for balanced path decomposition). The top-5 candidates are attempted in order, with VNFs assigned to the first feasible

node pair and spectrum allocated via first-fit on resulting segments.

2) Greedy Heuristic Baseline (WMSM-F):

WMSM-F [7] selects compute waypoints by jointly minimizing path length and segment imbalance $\delta_{\mathcal{W}}$: $\min_{\mathcal{W}} \left(\sum_{i=0}^{|\mathcal{W}|} d(w_i, w_{i+1}) + \lambda \cdot \delta_{\mathcal{W}} \right)$, where $\delta_{\mathcal{W}} = \max_i d(w_i, w_{i+1}) - \min_i d(w_i, w_{i+1})$, with first-fit spectrum allocation and blocking if no feasible configuration exists.

3) *Adapted VONE-DRL Baseline*: Inspired by Doherty *et al.* [8], who apply masked DRL to VONE over single-core EONs with separate node and link mapping agents. Both agents are adapted to the 4-VNF chain setting (the node agent selects the anchor compute node and the link agent handles per-segment RMCSA) and share the same hyperparameters and training budget as PPO-Full, ensuring a fair comparison under identical computational constraints.

IV. NUMERICAL EVALUATION

A. Simulation Setup

Simulations use a 24-node USNET topology with 42 bidirectional links (30–200 km), 7 cores per fiber, and 120 slots per core (840 slots/link). Spectrum follows one guard slot and core group search order $\{0, 2, 4\} \rightarrow \{1, 3, 5\} \rightarrow \{6\}$. Modulation is distance-adaptive: 16-QAM (≤ 500 km, 1 slot/Gbps), QPSK (≤ 1000 km, 2 slots/Gbps), and BPSK (≤ 2000 km, 4 slots/Gbps).

VNF hosting is enabled on 22 of 24 nodes, with each compute node having 150 compute units. Requests arrive as a Poisson process with $\lambda \in \{5, 10, \dots, 75\}$, exponential service times ($\mu = 0.5$), and each request specifies a bandwidth $b_{req} \sim \mathcal{U}(1, 20)$ Gbps and $F = 4$ VNF compute demands $f_i \sim \mathcal{U}(2, 10)$ normalized units. To stress-test performance under realistic traffic asymmetry, 30% of requests are hotspot traffic with source drawn uniformly from nodes $\{0, 5, 12\}$ and a random destination, while the remaining 70% use uniformly random source-destination pairs. Hotspot nodes $\{0, 5, 12\}$ were selected as geographically distributed high-degree nodes in the USNET topology, and the 30% hotspot fraction reflects traffic concentration patterns commonly observed in metro and backbone networks [24], where a small subset of nodes generates a disproportionate share of demand. Performance is evaluated over 10,000 requests per λ across 5 independent seeds.

Five strategies are evaluated: (1) *PPO-Full*: the proposed approach jointly selecting the compute node and the routing path, (2) *Endpoint-Biased*: distance-based heuristic ranking nodes by proximity to the source or destination, (3) *Equidistant*: distance-based heuristic ranking nodes by equidistance from source and destination, and (4) *WMSM-F*: the greedy waypoint-based baseline from [7], and (5) *VONE-DRL*: the decoupled RL baseline adapted from Doherty *et al.* [8]. All heuristic baselines apply first-fit spectrum allocation on the resulting path segments.

B. Results and Discussion

Fig. 1 shows the acceptance rate under 30% hotspot traffic. PPO-Full maintains the highest acceptance throughout, reaching $\approx 80\%$ at $\lambda = 75$, compared to $\approx 76\%$ for WSM-F, $\approx 74\%$ for Endpoint-Biased and VONE-DRL, and $\approx 70\%$ for Equidistant. PPO-Full's advantage widens progressively with load, confirming that joint node-path optimization adapts more effectively to hotspot-induced traffic concentration than deterministic strategies. VONE-DRL's low acceptance despite near-zero VONE blocking reveals that its decoupled design wastes spectrum resources on inefficient routing.

Fig. 3 shows RMCSA blocking. VONE-DRL performs worst ($\approx 26\%$ at $\lambda = 75$), confirming its decoupled link mapping agent makes spectrum-inefficient decisions. WSM-F achieves the lowest RMCSA blocking ($\approx 5\%$) through compact waypoint segments, while PPO-Full ($\approx 8\%$) outperforms Endpoint-Biased ($\approx 13\%$) and Equidistant ($\approx 18\%$).

Fig. 2 shows VONE blocking. VONE-DRL exhibits near-zero VONE blocking as its greedy chain placement always finds a feasible compute node, but at the cost of poor spectrum efficiency. WSM-F suffers the highest VONE blocking ($\approx 19\%$) as its distance-minimizing waypoint selection exhausts compute at hotspot-adjacent nodes, while PPO-Full achieves the lowest non-trivial VONE blocking ($\approx 12\%$), demonstrating that learned node selection adapts more effectively to concentrated hotspot demand.

Fig. 6 shows provisioning cost per VNR. VONE-DRL incurs the highest cost (≈ 85 at low load, declining to ≈ 71 at $\lambda = 75$), while PPO-Full maintains the lowest throughout (≈ 59 – 67). Equidistant remains consistently high (≈ 72 – 79) due to longer midpoint-biased segments, and WSM-F rises steeply at high load as it is forced onto longer waypoint paths under congestion. Fig. 4 shows VONE-DRL produces the highest fragmentation (≈ 0.14 at $\lambda = 75$), while PPO-Full achieves low fragmentation (≈ 0.06), comparable to Endpoint-Biased, and WSM-F achieves the lowest (≈ 0.05). However, WSM-F's lower fragmentation is an artefact of its higher blocking rate, that is, fewer accepted requests leave larger contiguous free spectrum regions, rather than reflecting any genuine spectrum-management advantage. Fig. 5 shows VONE-DRL reaches the highest utilization ($\approx 36\%$) despite lower acceptance, confirming excessive per-VNR spectrum consumption, while PPO-Full maintains $\approx 29\%$, comparable to WSM-F. PPO-Full's higher fragmentation and utilization relative to WSM-F reflect a deliberate trade-off: the RL agent accepts more requests by routing more aggressively, which produces higher fragmentation as a consequence of higher throughput rather than inefficient spectrum use.

V. CONCLUSION

This paper proposed PPO-Full, a PPO-based approach for network slice embedding in SDM-EONs that jointly optimizes compute node selection and RMCSA over K candidate nodes

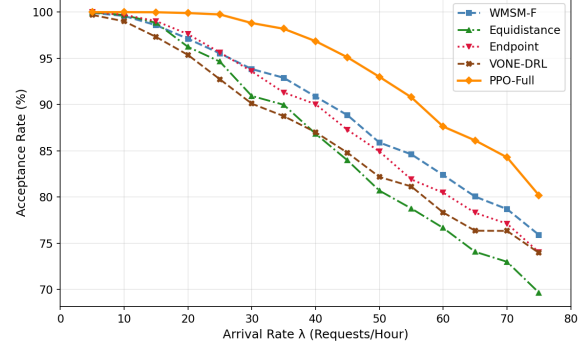


Fig. 1. Acceptance Rate vs. Arrival rates.

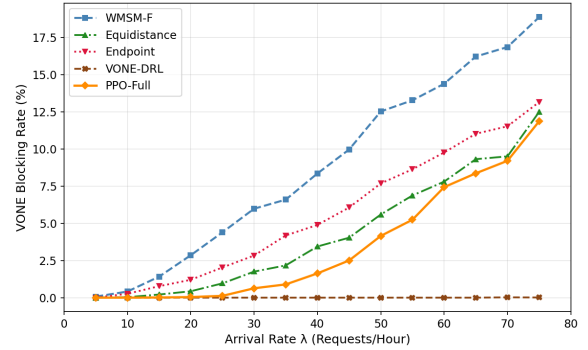


Fig. 2. Blocking Rate for Slice Embedding vs. Arrival rates.

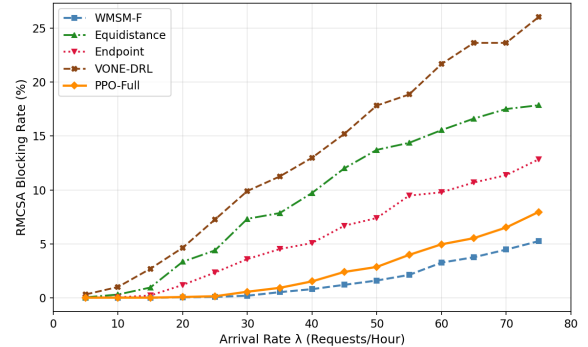


Fig. 3. Blocking Rate for RMCSA vs. Arrival rates.

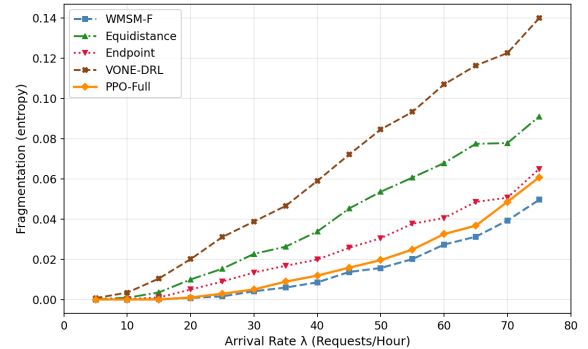


Fig. 4. Fragmentation vs. Arrival rates.

and paths selected via a multi-dimensional action space. Evaluated against Endpoint-Biased, Equidistant, WSM-F, and VONE-DRL [8] on a 24-node USNET topology under hotspot traffic, PPO-Full consistently achieved the highest acceptance

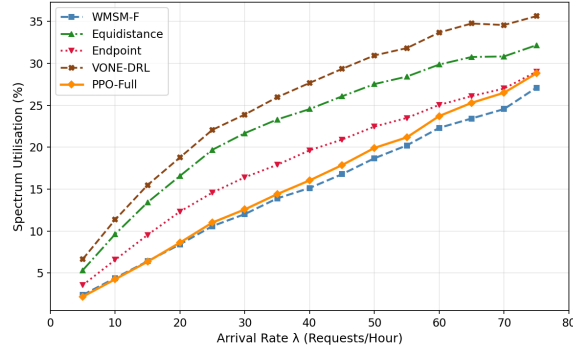


Fig. 5. Spectrum Utilization vs. Arrival rates.

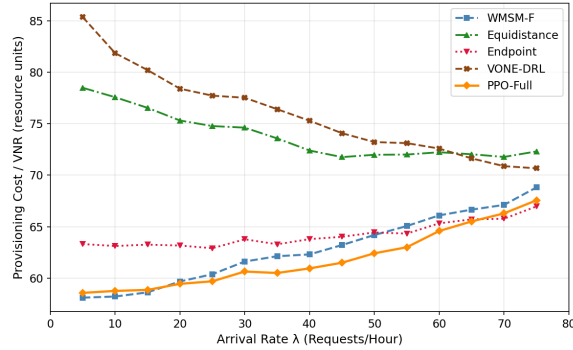


Fig. 6. Average Provisional Cost per VNR vs. Arrival rates.

and lowest per-VNR cost across all load levels. WMSM-F's lower fragmentation and RMCSA blocking are artefacts of conservative blocking rather than genuine spectrum efficiency, while VONE-DRL's decoupled design produced the worst RMCSA blocking despite near-zero compute blocking, confirming that joint compute-spectrum optimization is essential under realistic heterogeneous traffic. Future work will explore spectrum-aware path generation and removal of the K restriction through pointer networks or graph attention mechanisms.

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REFERENCES

- [1] B. C. Chatterjee, N. Sarma, and E. Oki, "Routing and spectrum allocation in elastic optical networks: A tutorial," vol. 17, no. 3, 2015. [Online]. Available: <https://doi.org/10.1109/COMST.2015.2431731>
- [2] Y. Wang and Q. Hu, "A path growing approach to optical virtual network embedding in slice networks," *Journal of Lightwave Technology*, vol. 39, no. 8, pp. 2253–2262, 2021.
- [3] L. Gong and Z. Zhu, "Virtual optical network embedding (vone) over elastic optical networks," *Journal of Lightwave Technology*, vol. 32, no. 3, pp. 450–460, Feb. 2014.
- [4] J. Gu, M. Zhu, Y. Wang, X. Cai, Y. Cai, J. Zhang, M. Lei, B. Hua, P. Gu, and G. Zhao, "Resource-efficient and qos guaranteed 5g ran slice migration in elastic metro aggregation networks using heuristic-assisted deep reinforcement learning," *Journal of Optical Communications and Networking*, vol. 15, no. 11, pp. 854–870, 2023.
- [5] D. Wang, W. Zhang, H. He, and C. Liu, "Node-fusion: Topology-aware virtual network embedding algorithm for repeatable virtual network mapping over substrate nodes," *IEEE Access*, vol. 9, pp. 142 581–142 592, 2021.
- [6] N. M. M. K. Chowdhury, M. R. Rahman, and R. Boutaba, "Virtual network embedding with coordinated node and link mapping," in *Proceedings of IEEE INFOCOM*, 2009, pp. 783–791.
- [7] D. Khanure, R. Gour, C. Li, and J. P. Jue, "Network slice embedding over space division multiplexed elastic optical networks," 2026. [Online]. Available: <https://arxiv.org/abs/2604.11936>
- [8] M. Doherty, Y. Zhang, and A. Beghelli, "Masked deep reinforcement learning for virtual network embedding on elastic optical networks," in *Proceedings of the 2023 International Conference on Optical Network Design and Modeling (ONDM)*. Coimbra, Portugal: IEEE, 2023.
- [9] Q. Zhang, X. Zhang, X. Gong, and L. Guo, "Crosstalk-avoid virtual optical network embedding over elastic optical networks with heterogeneous multi-core fibers," *Journal of Lightwave Technology*, vol. 40, no. 24, pp. 7687–7700, 2022.
- [10] T. Jin, S. Yin, and S. Huang, "A dynamic vone algorithm considering topology for hybrid services in sdm-eon," in *2023 Opto-Electronics and Communications Conference (OECC)*, 2023, pp. 1–6.
- [11] F. Tang, Y. Li, G. Shen, and G. N. Rouskas, "Minimizing inter-core crosstalk jointly in spatial, frequency, and time domains for scheduled lightpath demands in multi-core fiber-based elastic optical networks," *Journal of Lightwave Technology*, vol. 38, no. 20, pp. 5595–5607, 2020.
- [12] V. L. Y. N. S. Baljinder Singh Heera, Anjali Sharma, "Routing, modulation, core and spectrum assignment techniques for crosstalk management in multi-core fiber based spatially multiplexed elastic optical networks," *Photonic Network Communications*, vol. 48, pp. 35–50, 2024.
- [13] B. S. Heera, S. Petale, Y. N. Singh, and S. Subramaniam, "Rmcsa algorithm for congestion-aware and service latency aware dynamic service provisioning in software-defined sdm-eons," 2024. [Online]. Available: <https://arxiv.org/abs/2412.10685>
- [14] X. Chen, B. Li, R. Proietti, H. Lu, Z. Zhu, and S. J. B. Yoo, "DeepRMCA: A deep reinforcement learning framework for routing, modulation and spectrum assignment in elastic optical networks," *Journal of Lightwave Technology*, vol. 37, no. 16, pp. 4055–4067, 2019.
- [15] J. Pinto-Ríos, B. Dumas Feris, C. Vázquez, G. Saavedra, D. Bórquez-Paredes, N. Jara, R. Olivares, S. Amjad, A. Leiva, and C. Mas-Machuca, "Resource allocation in multicore elastic optical networks: A deep reinforcement learning approach," *Complexity*, vol. 2023, p. 4140594, 2023.
- [16] L. Xu, Y.-C. Huang, Y. Xue, and X. Hu, "Deep reinforcement learning-based routing and spectrum assignment of EONs by exploiting GCN and RNN for feature extraction," *Journal of Lightwave Technology*, vol. 40, no. 15, pp. 4945–4955, 2022.
- [17] V. Kumar, J. Halder, A. Mitra, E. Oki, and B. C. Chatterjee, "Inter-core and inter-mode crosstalk-avoided virtual network embedding in spectrally-spatially elastic optical networks," in *2022 IEEE 23rd International Conference on High Performance Switching and Routing (HPSR)*, 2022, pp. 125–130.
- [18] A. N. Zhang, L. and A. Khreishah, "Anycast planning in space division multiplexing elastic optical networks with multi-core fibers," *IEEE Communications Letters*, vol. 20, pp. 1983–1986, 2016.
- [19] P. Wright, M. C. Parker, and A. Lord, "Minimum- and maximum-entropy routing and spectrum assignment for flexgrid elastic optical networking," *IEEE/OSA Journal of Optical Communications and Networking*, vol. 7, no. 1, pp. A66–A72, 2015.
- [20] B. C. Chatterjee, N. Sarma, and E. Oki, "Routing and spectrum allocation in elastic optical networks: A tutorial," *IEEE Communications Surveys Tutorials*, vol. 17, no. 3, p. 1776–1800, 2015.
- [21] N. M. M. K. Chowdhury and R. Boutaba, "A survey of network virtualization," *Computer Networks*, vol. 54, no. 5, p. 862–876, 2010.
- [22] A. P. Bianzino, C. Chaudet, D. Rossi, and J. Rougier, "A survey of green networking research," *IEEE Communications Surveys Tutorials*, vol. 14, no. 1, p. 3–20, 2012, discusses NP-hardness of multi-resource allocation problems.
- [23] T. Tanaka, "Reinforcement-learning-based path planning in multilayer elastic optical networks [invited]," *Journal of Optical Communications and Networking*, vol. 16, no. 1, pp. A68–A77, 2024.
- [24] A. Akella, A. Bharambe, M. Reiter, and S. Seshan, "Detecting DDos attacks on ISP networks," in *Proceedings of the ACM SIGMOD Workshop on Management and Processing of Data Streams*, 2003.